

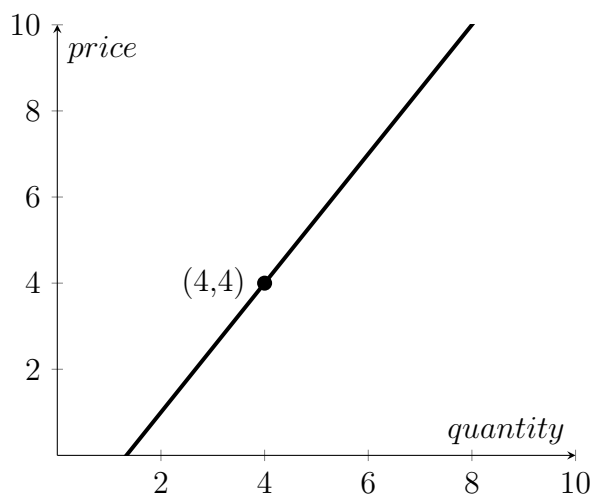
Practice with Shifting Curves

ECON 2010 – Macroeconomics
Spring 2019

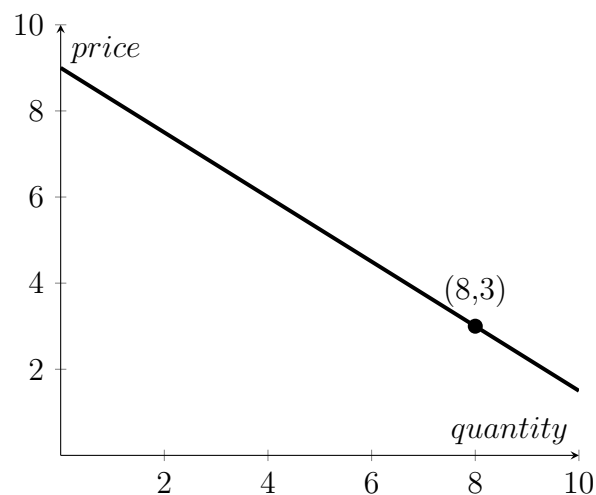
Practice Problem 1

Figure 1: Market supply, market demand and equilibrium for dog bones

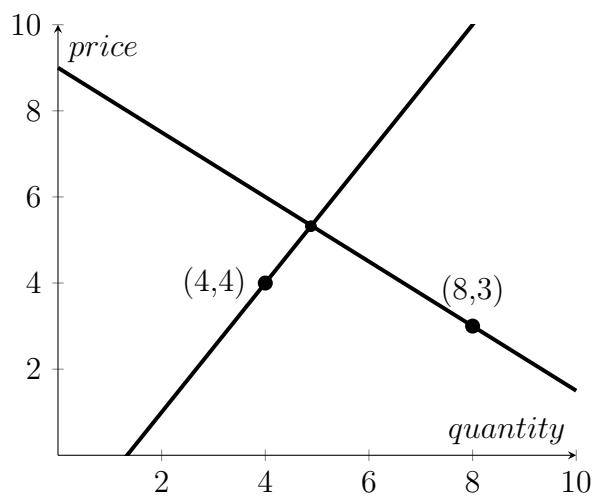
Plot A: Market supply curve



Plot B: Market demand curve



Plot C: Market supply and market demand, a.k.a. market equilibrium



Consider the market for dog bones for pets in the United States. (Notice that this is a market for *goods* rather than a market for *labor*.) Let's assume that there are many manufacturers (firms who produce) of dog bones. And because this is just one market, dog bones are indistinguishable across manufacturers. To make this market seem more like the real world, we won't even put a number on how many manufacturers there are.

Even though dogs eventually consume the dog bones, it's human beings (dog owners) in the US who demand dog bones. The number of individual consumers is quite large — on the order of millions of people, but we won't say here exactly how many, again, to make this more like a real-world market. We know that there are a lot of individuals who make up demand, but we don't always know exactly how many across an entire market, especially a large one like for dog bones.)

In Figure 1, I've drawn the market supply curve for dog bones in Plot A, the market demand curve for dog bones in Plot B, and the market equilibrium that puts the market demand and market supply curves together on one plot (Plot C). The point at which the two curves cross in Plot C is *approximately* (5,5). (It's not quite there, but that's close enough for our purposes.)

NOTE: The price in all of the plots is measured in *dollars* per unit (i.e. price per dog bone). The quantity in all of the plots is measured in *millions of dog bones*. So, a 2 on the quantity axis represents *2 million dog bones*.

Given this information about the market, answer the following questions.

(a) Who is a supplier in this market? Who is providing dog bones to the market?

(b) Who is a demander in this market? Who is purchasing dog bones from the market?

(c) In **one or at most two complete sentences**, state what is represented by the point (4,4) in Plot A (the market supply curve). Be thorough in your answer.

- (d) In **one or at most two complete sentences**, state what is represented by the point (8,3) in Plot B (the market demand curve). Be thorough in your answer.
- (e) Given this market equilibrium in Plot C, explain briefly why a consumer who wants to buy a dog bone at \$4 (i.e. who demands a dog bone at \$4) cannot buy a dog bone at that price. Hold all else fixed when you answer this question.
- (f) Given the market equilibrium in Plot C, explain briefly why a firm/producer who wants to sell a dog bone at \$6 (i.e. who supplies a dog bone at \$6) cannot sell a dog bone at that price. Hold all else fixed when you answer this question.

(g) Given the market equilibrium in Plot C, consider the following change in the market. Consider that a million new people move to the US. Most of these people are dog owners who feed their dogs bones. Answer the following questions about this change to the market. Hold all else fixed when you consider this change and how it affects the market.

(1) Which curve or curves will shift as a result of this market change? You could say **market supply**, **market demand**, **none** or **both**.

(2) In which direction will the curve or curves shift that you indicated would shift in part (e1)? For each you can say **right** or **left**. If no curves shift, just write “none” here.

(3) In just a few sentences, explain briefly **why** the curve or curves shift in the direction that you indicated they would shift. If no curves shift, just write “none” here.

- (4) Draw a new plot that indicates the curve shift or shifts that you thought would take place. **Draw** the starting curves and equilibrium from Plot C with solid lines, and **draw** any curve shifts with dashed lines. **Indicate** the starting equilibrium from Plot C with a point and label it **A**. **Indicate** the new equilibrium according to your curve shifts with a point and label it **B**. If no curves shift, just re-draw Plot C.

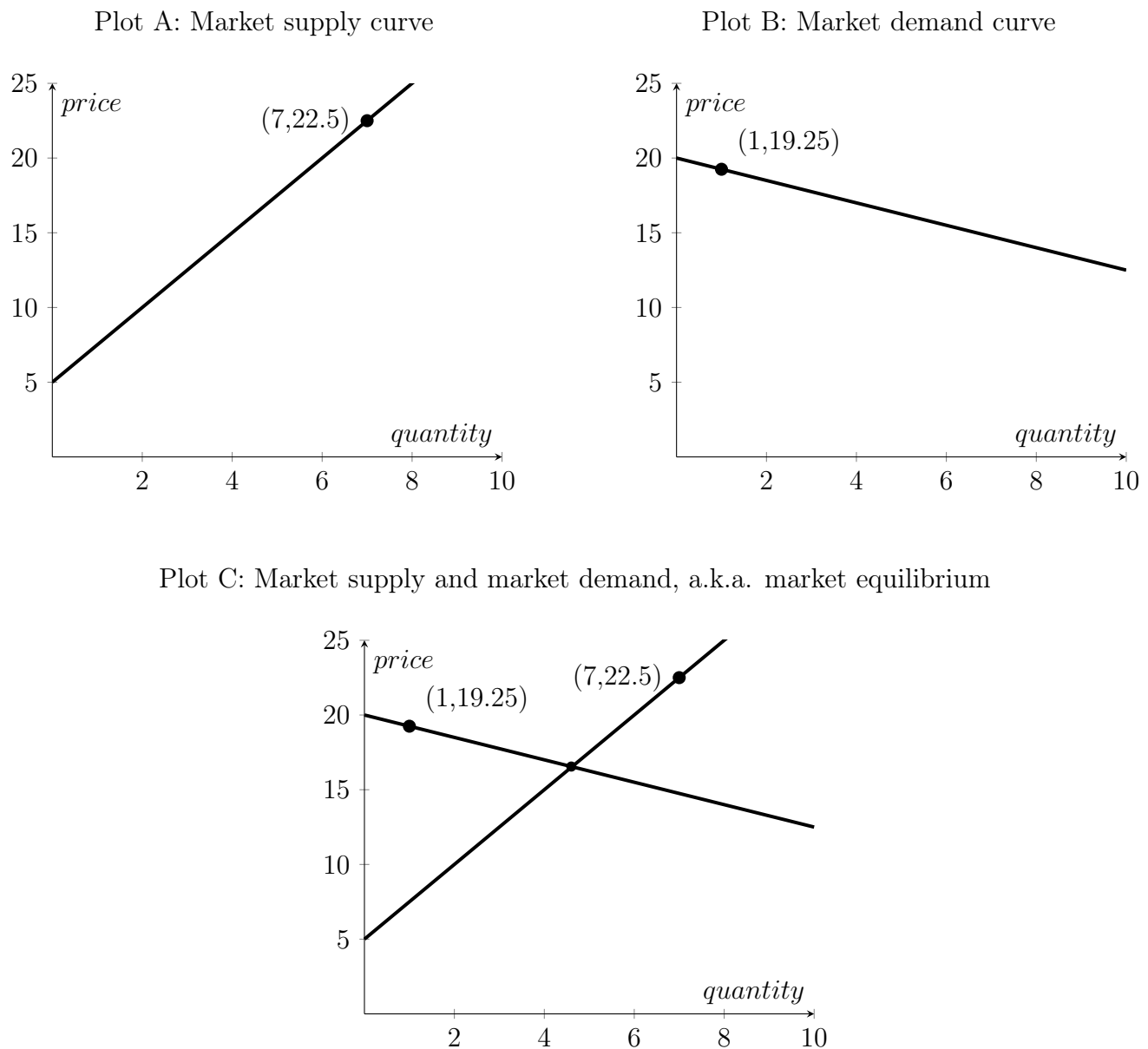
(Just draw the shifted curves well enough to visualize any equilibrium change that may have occurred. I do not give you enough information to know by how much the curves shifted, if at all.)

- (5) According to the new equilibrium, how have **each** price and quantity changed (if at all). For each price and quantity you can say that they have gone **up**, **down**, **stayed the same**, or the change is **not possible to know** given the information provided. (Remember: An equilibrium is a price **and** a quantity. You cannot correctly answer this question by just saying “down” or “up.” An equilibrium can’t go “down” or “up.” *What* went down; *what* went up?)

Practice Problem 2

Consider the labor market for drywall installers in the United States. As we did in Problem 3 above, let's assume that we don't know exactly how many firms there are in the US that hire drywall installers or exactly how many individual laborers there are in the market. We'll just say that there are many many firms, and many many individual laborers. Assume that all drywall installers are indistinguishable to firms, and all drywall firms are indistinguishable to drywall installers (this is an example of a simplifying assumption, but it's also an assumption that's part of our definition of a market).

Figure 2: Market supply, market demand and equilibrium for drywall installers



In Figure 1, I've drawn the labor market supply curve for drywall installers in Plot A, the labor market demand curve for drywall installers in Plot B, and the market equilibrium that puts the market demand and market supply curves together on one plot (Plot C). The point at which the two curves cross in Plot C is *approximately* (4.6,16.5). (It's not quite there, but that's close enough for our purposes.)

NOTE: The price (i.e. the wage) in all of the plots is measured in *dollars per hour* of work (i.e. it's an hourly wage). The quantity in all of the plots is measured in *millions of total hours* worked. So, a 2 on the quantity axis represents 2 million total hours worked.

As you answer this question, assume (simplifying assumption) that all workers and all firms are located in exactly the same place. That is, don't try to complicate the problem by thinking about where workers are located geographically.

Given this information about the market, answer the following questions.

(a) Who is a supplier in this market? That is, who is providing drywall labor to this market?

(b) Who is a demander in this market? That is, who is demanding drywall labor in this market?

(c) In **one or at most two complete sentences**, state what is represented by the point (7,22.5) in Plot A (the labor market supply curve). Be thorough in your answer.

(d) In **one or at most two complete sentences**, state what is represented by the point (1,19.25) in Plot B (the labor market demand curve). Be thorough in your answer.

(e) Given this market equilibrium in Plot C, explain briefly why a consumer who wants to earn a wage of \$18 per hour (i.e. who supplies their labor at the hourly wage of \$18) cannot find hours to work at that wage. Hold all else fixed when you answer this question.

(f) Given the market equilibrium in Plot C, explain briefly why a firm/producer who wants to hire a drywall installer at \$15 (i.e. who demands drywall installation at \$15) cannot buy labor at that price. Hold all else fixed when you answer this question.

(g) Given the market equilibrium in Plot C, consider the following changes in the market. **First**, consider an increase in US immigration. This new population needs houses, which have drywall in them. Therefore the price that drywall installation firms can charge homebuilders to install drywall goes up. (Note: This market change is **not in and of itself** a change in the wage for a drywall installer. It is a change in the price of the *output* that drywall installation *firms* provide. Drywall installers are just *one* input for drywall installation firms. So, I'm not saying that the market change is a change in the wage; I'm saying that something we usually hold fixed when we consider changes in wages — i.e. the price of the output product for which the labor we're considering as an input — has changed.) Assume that *none* of the new immigrants can install drywall.

Second, in addition to the first change, consider a massive shortage of nail guns in the US. Nail guns are an important tool used by drywall installers to install drywall. Without nail guns, to install the same amount of drywall as before requires drywall installers to hammer nails in by hand, which takes much longer. Assume drywall installers have no preference about whether they use a nail gun or hammer nails. That is, they are completely indifferent about which tool they use.

Answer the following questions about this change to the market. Hold all else fixed when you consider these change to the market. **(15 points total)**

(1) Which curve or curves will shift as a result of these market changes? You could say **labor market supply**, **labor market demand**, **none** or **both**.

(2) In which direction will the curve or curves shift that you indicated would shift in part (e1)? For each you can say **right** or **left**. If no curves shift, just write “none” here.

(3) In just a few sentences, explain briefly **why** the curve or curves shift in the direction that you indicated they would shift. If no curves shift, just write “none” here.

(4) Draw a new plot that indicates the curve shift or shifts that you thought would take place. **Draw** the starting curves and equilibrium from Plot C with solid lines, and **draw** any curve shifts with dashed lines. **Indicate** the starting equilibrium from Plot C with a point and label it **A**. **Indicate** the new equilibrium according to your curve shifts with a point and label it **B**. If no curves shift, just re-draw Plot C.

(This is not a drawing test; just draw the shifted curves well enough to visualize any equilibrium change that may have occurred. I do not give you enough information to know by how much the curves shifted, if at all.)

- (5) According to the new equilibrium, how have **each** price (i.e. the wage) and quantity changed (if at all). For each price and quantity you can say that they have gone **up**, **down**, **stayed the same**, or the change is **not possible to know** given the information provided. (Remember: An equilibrium is a price **and** a quantity. You cannot correctly answer this question by just saying “down” or “up.” An equilibrium can’t go “down” or “up.” *What* went down; *what* went up?)

Tips and Things to Remember

1. Ask yourself whether the market is a market for goods or a market for labor. If it's a goods market, those who are supplying the good will be firms who produce the good. Those who demand the good will be consumers who are buying it. If it's a labor market, those who supply to the market will be laborers, while those who demand will be those who need to use that labor. If you work in the wood plant, the plant/the state demands labor, while you, as a worker, supply your labor.
2. Because labor markets and goods markets have different suppliers and different demanders, *how* a change effects them will be different. Therefore, we *cannot* say that a given change will *always* shift the demand curve right. Instead, *whether the demand curve shifts will depend on what specific market we are talking about.*
3. How do we come up with the demand curve for an individual demander in a market? The best way to think about how we do this is to imagine a survey that goes out to the demander. The survey asks the demander how much of the thing (labor, good, service) that demander *would* want *were* the price to be a particular price. We ask them this question for a lot of hypothetical prices.
4. Similarly, for the supply curve, we survey all suppliers in the market and ask them how much of the thing (good, labor, service) they would be willing to supply were the price a particular price. And we ask them this question for a lot of hypothetical prices.
5. To arrive at market demand, we add up all of the quantities hypothetically demanded by all of the demanders in the market at every given price. To arrive at market supply, we add up all of the hypothetical quantities supplied by suppliers in the market at every given price.
6. To know the equilibrium price and quantity in the market, it is not enough to know *just* market supply or *just* market demand. It is not enough to know how much all suppliers in the market, for instance, *would* supply *were* the price to be a particular price. Nor is it enough to know how much all demanders *would* demand were the price a particular price. Intuitively you may find this makes sense. If we want to know how many iPhones will be sold in 2019, we can't just know how many iPhones Apple is willing to produce at different prices. Nor can we just know how many iPhones consumers are willing to buy at different prices. Instead, we need to know both so that we can see the price and quantity that make both suppliers and demanders happy. This equilibrium price and quantity is the combination where neither suppliers nor demanders can be better off if they unilaterally asked for/demanded a higher/lower price.
7. Ask yourself what a given point on a curve represents. When we plot a line or a curve, that line or curve represents the relationship between whatever is quantified on the x axis and what is quantified on the y axis. In market curves, the x axis is always some representation of quantity, while the y axis is some representation of price. If it is a labor market, we have a particular name for the price of labor, which is "the wage."
8. Everything that is not represented on one of the two axes of a plot is *held fixed*. So, on a demand curve, when we consider how a change in price changes the hypothetical quantity demanded, we do that holding every other thing you could possibly imagine fixed. Because every other thing you could possibly imagine is NOT one of the two axes on the plot. The only thing that is different between one quantity on the plot and another quantity, is how the price (the other axis) has changed.
9. When you're given something that is NOT one of the two axes and asked whether it shifts a

curve, ask yourself: “At a given price, does this change mean that demanders in the market should demand more, less or the same?” Then ask yourself: “At a given price, does this change mean that suppliers in the market should supply more, less or the same?” You may find it helpful to *imagine* yourself as a demander in the market, and to imagine yourself as a supplier. If you were a demander, would this change mean that you would answer the survey about what you demand at a given price differently? If you were a supplier, would this change mean that you would answer the survey about what you would supply at a given price differently? If so, how?

10. Example: In the US goods market for couches, those who supply couches are furniture firms. Those who demand couches are individuals/households. What happens if more people move to the US, people who we assume don’t know how to make couches, but who we will assume do buy couches. Does market demand shift? Yes. At any given price, there are now more people who want to buy couches, thus more demand at that price. Even if only 1 of a million people who moved to the US wanted a couch, that one person would contribute demand. The curve shifts to the right, because at any given price, the quantity demanded (what is represented on the x axis, has increased). Does market supply shift? No. At any given price, firms who make couches don’t care how many people are demanding couches. The market demand curve does NOT represent how many couches firms want to *sell* or can sell, it represents how many they are willing to produce/supply at a given price. Their willingness to produce or supply is a function of how much it costs to make a couch — the cost of labor (wages), the cost of labor relative to its quality (skill), the cost of regulatory compliance, the cost of raw materials, the cost of machinery to make couches, etc. So the supply curve does not shift when there is a population increase.
11. Example: In the US market for laminate flooring labor, those who supply labor are individuals like you and me, and those who demand labor are firms who make laminate flooring (like the wood plant). What happens if every high school student in the US gets six weeks of training in how to make laminate flooring? Assume that these skills are *only* useful for making laminate flooring. Does market demand shift? Remember, demanders are those firms who make flooring and need people who can operate machinery and do all of the flooring jobs. Yes, demand does shift. At any given price, labor is now more skilled. *All else equal*, the productivity of labor has increased relative to other inputs to production, like capital (i.e. machines). So, at a given wage (i.e. price of labor), firms prefer to hire more of these more skilled workers. You can think of yourself as the owner of a firm who makes flooring. If the hypothetical wage is \$15 per hour, do you want to hire more skilled workers than you would at the same wage if they were less skilled? Yes. Now, does market supply shift? No. At a given wage, nothing that affects how much laminate floor laborers want to work has changed. They have more skills, which are useful to firms, but don’t make a difference in how much you want to do this job at a given wage.